

SVD

Kachun Huang

2025-03-27

SVD in R

Generalized Inverse

Consider the following system of linear equations:

$$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Since

$$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$$

is a Singular matrix, where the determinant is 0. So there is no exact solution but it is possible to get a solution in the least-squares sense.

Recall, the definition of inverse, given A is non-singular:

$$AA^{-1}A = A$$

Now suppose A is singular, then the inverse A^{-1} doesn't exist.

But, we can compute a Moore-Penrose generalized Inverse A^+ , such that

$$AA^+A = A$$

where A^+ is given by

$$A^+ = (A^T A)^{-1} A^T$$

Relation to SVD, Moore-Penrose generalized Inverse A^+ is given by

If $A = U\Sigma V^T$, then $A^+ = V\Sigma^+U^T$

where,

- U is a $n \times p$ orthonormal matrix
- V is a $p \times p$ orthonormal matrix
- Σ is the $p \times p$ diagonal matrix

- Σ^+ is the Σ diagonal matrix but take recipirical of all diagonal entries and they have the same rank

```
X <- matrix(
  c(2,4,1,2),
  nrow=2
)
X
```

```
##      [,1] [,2]
## [1,]    2    1
## [2,]    4    2
```

Do SVD of X

```
svd_X <- svd(X)
svd_X
```

```
## $d
## [1] 5.000000e+00 1.040614e-16
##
## $u
##      [,1] [,2]
## [1,] -0.4472136 -0.8944272
## [2,] -0.8944272  0.4472136
##
## $v
##      [,1] [,2]
## [1,] -0.8944272 -0.4472136
## [2,] -0.4472136  0.8944272
```

```
U <- svd_X$u
Sigma <- diag(svd_X$d)
V <- svd_X$v
```

Find Σ^+

```
Sigma_plus <- diag(c(1/svd_X$d[1],0))
Sigma_plus
```

```
##      [,1] [,2]
## [1,]  0.2    0
## [2,]  0.0    0
```

Find X^+

```
Xinv <- V %*% Sigma_plus %*% t(U)
Xinv
```

```
##      [,1] [,2]
## [1,] 0.08 0.16
## [2,] 0.04 0.08
```

We can also use `ginv()` from MASS Library

```
library(MASS)
ginv(X)
```

```
##      [,1] [,2]
## [1,] 0.08 0.16
## [2,] 0.04 0.08
```

So with the A^+ , we can compute the solution of this system of linear equation

$$x = A^+ y$$

```
Xinv %*% c(1,1)
```

```
##      [,1]
## [1,] 0.24
## [2,] 0.12
```

Least Square Estimation

Consider the multiple linear regression model in matrix notation:

$$Y = \beta X$$

Then we can derive estimation of the β using Moore-Penrose generalized Inverse

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Example

```
set.seed(123)
n <- 100
x1 <- rnorm(n)
x2 <- rexp(n)
y <- 2-1*x1+3*x2+rnorm(n)

X <- cbind(1,x1,x2)
beta.est <- solve(t(X) %*% X) %*% t(X) %*% y
beta.est
```

```
##      [,1]
##      2.0608099
## x1 -0.8159223
## x2  2.9271224
```

OLS summary

```
fit <- lm(y ~ X)
summary(fit)
```

```
##
## Call:
## lm(formula = y ~ X)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3625 -0.8192  0.0136  0.7848  2.4862
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.0608     0.1627  12.668 < 2e-16 ***
## X              NA           NA      NA      NA
## Xx1          -0.8159     0.1188  -6.868 6.22e-10 ***
## Xx2           2.9271     0.1159  25.248 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.078 on 97 degrees of freedom
## Multiple R-squared:  0.8779, Adjusted R-squared:  0.8754
## F-statistic: 348.8 on 2 and 97 DF,  p-value: < 2.2e-16
```